RAMDEOBABA UNIVERSITY, NAGPUR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.Tech. CSE (Data Science and Analytics)



SEMESTER V

LAB

PROJECT REPORT

Weather And Climate Insights

Submitted by

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Weather And Climate Insights

DST Lab Project Report

1. PROJECT OBJECTIVE

The objective of this project is to build an interactive weather and climate dashboard in Tableau using historical daily weather observations for major Indian cities over the period 2000–2024. The dashboard is intended to make weather patterns easier to explore through visual summaries, city comparisons, rainfall analysis, and long-term temperature trends.

- Load and explore the selected daily weather dataset.
- Review the date, city, apparent temperature, precipitation, and wind-speed fields and prepare them for visualization.
- Create KPI worksheets and multiple charts to compare seasonal, city-wise, and yearly patterns.
- Combine the worksheets into an interactive dashboard with a consistent visual style.
- Present historical patterns clearly while recognizing that historical visualization alone is not a weather forecast or proof of climate change.

2. INTRODUCTION

Weather data contains useful information about temperature, rainfall, wind, and how these conditions vary by place and time. However, reviewing a large table of daily observations can make it difficult to identify seasonal differences or compare cities. Data visualization helps convert these observations into charts that are easier to interpret.

This project uses Tableau to create a Weather and Climate Insights dashboard for historical observations from major Indian cities. The dashboard combines KPI cards, line charts, bar charts, a map, a heatmap, and a scatter plot. Users can visually compare weather conditions and investigate patterns over time.

2.1 Problem Statement

The main problem addressed is the difficulty of interpreting daily weather records and comparing weather conditions across several cities and years from raw tabular data alone. The project provides a single visual interface for exploring temperature, rainfall, wind speed, and city-wise differences.

3. PROJECT METHODOLOGY

3.1 Dataset Loading and Exploration

Dataset used: India Daily Weather (2000–2024) – Major Cities. The project work describes daily weather observations for multiple Indian cities. The dashboard indicates coverage of 10 cities and includes date and city dimensions, apparent maximum and minimum temperature measures, precipitation, and wind speed.

The exact download URL and the dataset's total row and column counts were not provided in the project materials available for this report. These details should be copied from the original dataset source and the Tableau Data Source page before final submission; they have not been estimated here.

Fields used in the dashboard:

- Date: used to group observations by month, day, and year.
- City: used to compare weather observations between cities and to plot locations on the map.
- Apparent maximum temperature: used in city comparisons, the monthly trend, and the yearly trend.
- Apparent minimum temperature: used alongside apparent maximum temperature in the monthly trend.
- Precipitation: used for monthly rainfall totals and the daily rainfall heatmap.
- Wind speed: used for the wind-speed KPI and the temperature-versus-wind-speed scatter plot.

3.2 Data Preprocessing

The data was used as the basis for Tableau visualizations. The relevant fields were checked for suitable data roles, such as Date for time-based analysis, City for categorical comparisons, and numeric measures for temperature, precipitation, and wind speed. Date fields were grouped into calendar periods for monthly and yearly charts.

The report does not claim a particular number of removed rows, missing values, or duplicates because those counts are not visible in the supplied screenshot. Before submission, verify missing values and duplicates in the source data and document any cleaning actually performed. If no records were changed, state that the dataset was used without row-level cleaning.

4. PROJECT IMPLEMENTATION

4.1 Tools and Technologies

Tool / Technology	Version	Purpose
Tableau Desktop	Latest version available during implementation; exact build not shown	Data visualization, worksheet creation, dashboard assembly, formatting, and interactive analysis.
India Daily Weather dataset	2000–2024 coverage stated in project description	Historical weather observations for city-wise and time-based analysis.

4.2 Step-wise Implementation

- Opened Tableau Desktop and connected to the selected weather dataset.

- Reviewed the available fields and assigned appropriate data roles to date, city, and numeric weather measures.
- Created a monthly apparent-temperature line chart comparing the average maximum and minimum apparent temperatures.
- Created a monthly rainfall bar chart using precipitation aggregated by month.
- Created a city-wise temperature comparison chart to compare average maximum apparent temperature between cities.
- Created a map view to show city-wise temperature and rainfall information geographically.
- Created a city-by-month temperature heatmap to highlight differences in monthly temperature patterns.
- Created a scatter plot to explore the relationship between apparent temperature and wind speed, with city represented by color.
- Created a daily rainfall calendar/heatmap to inspect precipitation by date.
- Created a yearly average maximum apparent-temperature trend to examine changes across years.
- Created KPI worksheets for average maximum apparent temperature, total rainfall, average wind speed, and number of cities.
- Assembled the worksheets into a dashboard, added a title and light weather-themed background/banner, and arranged the views for comparison.

4.3 Dashboard Design

The dashboard is titled “WEATHER & CLIMATE INSIGHTS” with the subtitle “India / Historical Weather Analysis / 2000–2024.” A light blue and warm pastel background is used to create a weather-themed appearance. KPI cards are placed near the top, followed by charts arranged in rows. The dashboard screenshot shows the worksheet views together on a single Tableau dashboard with a custom size of 1400 × 900 pixels.

The supplied screenshot shows four KPI areas and the following analytical views: monthly average maximum and minimum apparent temperature, total rainfall by month, average maximum temperature by city, city-wise temperature and rainfall map, city-wise monthly temperature pattern, temperature versus wind speed, daily rainfall calendar/heatmap, and yearly average maximum temperature.

5. CALCULATED FIELDS AND AGGREGATIONS

Tableau uses calculated fields and aggregations rather than DAX, which is associated with Power BI. The dashboard can be built using the following measure aggregations and date groupings, provided they match the actual field names in the workbook:

- Average maximum apparent temperature: AVG([Apparent Temp Max]) or the equivalent maximum apparent-temperature field in the dataset.

- Average minimum apparent temperature: AVG([Apparent Temp Min]) or the equivalent minimum apparent-temperature field.
- Total rainfall: SUM([Precipitation]). Confirm that precipitation is recorded in millimetres before displaying the unit mm.
- Average wind speed: AVG([Wind Speed]) or the exact wind-speed field name.
- Cities covered: COUNTD([City]).
- Monthly analysis: group the Date field by Month; yearly trend: group Date by Year; daily rainfall: use the exact Date field.

These are recommended expressions based on the dashboard's purpose. Replace field names with the exact names in the Tableau Data pane. If the workbook already uses different aggregations, report the calculations actually used. Check the Total Rainfall KPI worksheet to ensure it displays only the intended precipitation measure.

6. RESULTS AND DISCUSSION

The implemented dashboard brings several views of historical weather data into one interface. The monthly temperature line chart supports comparison of apparent maximum and minimum temperatures across the calendar year. The rainfall bar chart highlights months with comparatively higher or lower precipitation. The city comparison chart supports comparison between cities, while the map provides a geographic view where locations are correctly resolved. The heatmaps show patterns across dates and months, and the scatter plot allows visual inspection of temperature and wind-speed relationships.

The supplied screenshot visibly reports an average maximum apparent-temperature KPI of 33.06, an average wind speed KPI of 15.52, and 10 cities covered. The rainfall KPI card appears to display more than one numeric value in the screenshot; therefore, its final value should be checked in Tableau before the report is submitted. These displayed values are dashboard outputs, not independently validated climate statistics.

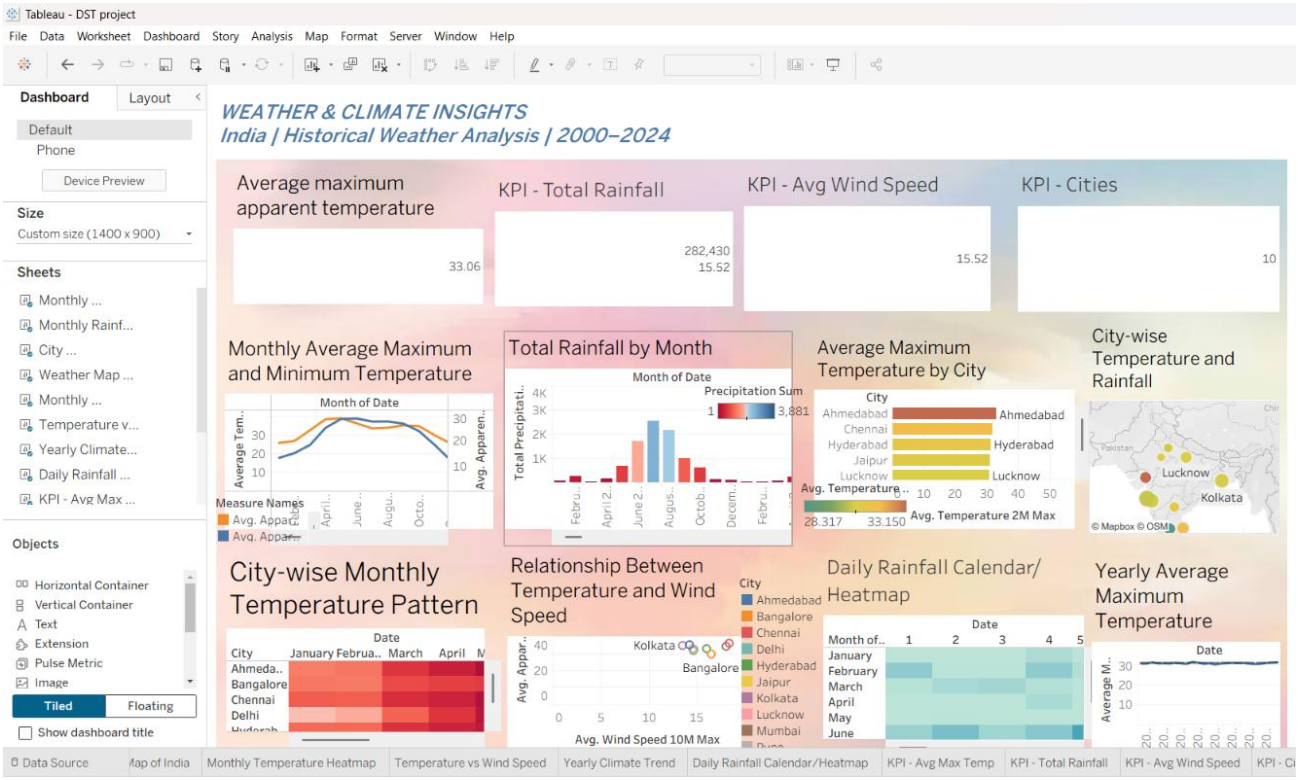


Figure 1. Weather and Climate Insights dashboard implemented in Tableau (screenshot supplied by the project team).

6.1 Discussion and Limitations

- The dashboard is descriptive: it summarizes historical observations and does not predict future weather.
- Apparent temperature is not the same as measured air temperature; chart titles and units should consistently identify it as apparent temperature.
- The map depends on correct geographic roles and location resolution in Tableau.
- The source URL, exact dataset size, missing-value counts, duplicate counts, and Tableau build number should be recorded from the actual files/software before final submission.
- Historical trends in this dashboard alone are not sufficient to establish a formal climate-change conclusion; longer-term analysis and appropriate statistical methods would be needed.

7. CONCLUSION

The Weather and Climate Insights project demonstrates the use of Tableau to turn historical daily weather observations into an interactive visual dashboard. It combines KPI summaries with monthly, daily, city-wise, geographic, and yearly views of apparent temperature, precipitation, and wind speed. The dashboard makes it easier to compare cities and inspect seasonal patterns. Future improvements may include verified data-quality documentation,

interactive filters across all relevant sheets, improved map location matching, clearer KPI formatting, and a separate detailed-analysis dashboard.

8. GITHUB LINK

https://github.com/Aashi-Kothale/Weather-and-Climate-Insights/tree/main

9. REFERENCES

[1] India Daily Weather (2000–2024) – Major Cities dataset. Insert the exact dataset publisher and download URL used by the project team.

[2] Tableau Help — Build a Dashboard: https://help.tableau.com/current/pro/desktop/en-us/dashboards_create.htm

[3] Tableau Help — Create a Map: https://help.tableau.com/current/pro/desktop/en-us/maps_howto.htm

[4] Tableau Help — Create Calculated Fields: https://help.tableau.com/current/pro/desktop/en-us/calculations_calculatedfields_create.htm